Sound speed $c \approx 340$ m/s (Air) 1500 m/s (Water)

Threshold of ear: $10^{-10} \le p'/p_0 \le 10^{-3}$

PWL (Power level) $= 10\log_{10}\left(\frac{\text{Sound power output}}{10^{-12}\ W}\right)$

$= 10\log_{10}(\text{Sound power output}) + 120\ dB$

SPL (Sound Pressure level) $= 20\log_{10}\left(\frac{\sqrt{\overline{p'^2}}}{2\times10^{-5}\ N/m^2}\right)$

$p'^2_{rms} = \overline{p'^2}$

threshold of $\begin{cases} \text{ear} = 2\times10^{-5}\ N/m^2 = 0\ dB \\ \text{pain} = 10^2\ N/m^2 = 134\ dB \end{cases}$

IL (Sound Intensity Level) $= 10\log_{10}\left(\frac{\bar{I}}{10^{-2}\ W/m^2}\right)$

$= 10\log_{10}\bar{I} + 120\ dB$

1-D wave equation (linearised)

Mass: $\frac{\partial \rho'}{\partial t} + \rho_0\, \text{div}\, v = 0$ $\xrightarrow{\frac{\partial}{\partial t}}$

Momentum: $\rho_0 \frac{\partial v}{\partial t} + \text{grad}\, p' = 0$ $\xrightarrow{\text{div}}$

wave equation $\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0$

isentropic, $p' = c^2 \rho'$

$\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = 0$

solution $\rightarrow p'(x,t) = f(x-ct) + g(x+ct)$

Harmonic $p'(x,t) = Ae^{i\omega(t-x/c)} + Be^{i\omega(t+x/c)}$

wave speed $c^2 = \gamma R T$, $R = 287\ J/(kg\cdot K)$

Acoustic partical velocity (限于平面波) $\begin{cases} p' = \rho_0 c u & (+) \\ p' = -\rho_0 c u & (-) \end{cases}$

Acoustic potential velocity

$u = \frac{\partial \phi}{\partial x}$, $p' = -\rho_0 \frac{\partial \phi}{\partial t}$

Suppose that $\phi = e^{i\omega t - ikx}$

$\Rightarrow p' = -i\omega\rho_0\phi$, $u = ik\phi$

Energy for 1-D:

$\begin{cases} \text{Mass eq.} \times p'c^2 \\ \text{Momentum eq.} \times u \end{cases} \Rightarrow \frac{\partial}{\partial t}\left(\frac{1}{2}\rho_0 u^2 + \frac{c^2\rho'^2}{2\rho_0}\right) + \frac{\partial}{\partial x}(p'u) = 0$

$\frac{\partial}{\partial t}(e_p + e_k) = -\nabla I = -\frac{\partial I_i}{\partial x_i}$

Sound spectra

$S(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(\omega) e^{i\omega t}\, d\omega$

$F(\omega) = \int_{-\infty}^{+\infty} S(t) e^{-i\omega t}\, dt$

$\int_{-\infty}^{+\infty} e^{-i\omega t} S(t-t_0)\, dt = e^{-i\omega t_0}$

$S(t-t_0) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{i\omega(t-t_0)}\, d\omega$

$\int_{-\infty}^{+\infty} e^{ixy}\, dx = 2\pi\delta(y)$

3-D wave eq.

near a singularity $\phi \sim r^{-\alpha}$ $(\alpha > 0)$

$\Rightarrow \nabla^2\phi \sim r^{-\alpha-2}$, $\frac{1}{c^2}\frac{\partial^2\phi}{\partial t} \sim \frac{\omega^2}{c^2} r^{-\alpha}$

球坐标形式下的波动方程

$\nabla^2 p' = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial p'}{\partial r}\right) + \underbrace{\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial p'}{\partial\theta}\right) + \frac{1}{r^2\sin\theta}\frac{\partial^2 p'}{\partial\varphi^2}}_{\text{neglect}}$

$\Rightarrow \frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial p'}{\partial r}\right) = 0$

$\hookrightarrow \frac{1}{c^2}\frac{\partial^2}{\partial t^2}(rp') - \frac{\partial^2}{\partial r^2}(rp') = 0$

$\Rightarrow rp' = f(r-ct) + f(r+ct)$

Solution in 3-D field

$p'(r,t) = \frac{A}{r}e^{i\omega(t-r/c)}$ $\xrightarrow{\rho_0\frac{\partial u}{\partial t} = -\frac{\partial p'}{\partial r}}$

$i\omega\rho_0 u = \left(\frac{A}{r}\frac{i\omega}{c} + \frac{A}{r^2}\right)e^{i\omega(t-r/c)}$

far field / near field

$\Rightarrow u(r,t) = \frac{1}{\rho_0 c}\sqrt{1+\left(\frac{c}{\omega r}\right)^2}\, e^{-i\varphi}\frac{A}{r}e^{i\omega(t-r/c)}$, $\varphi = \tan^{-1}\frac{c}{\omega r}$

p', u not in phase, $\begin{cases} p' \text{ lead } 90^\circ, & \text{near field} \\ \text{nearly in phase}, & \text{far field} \end{cases}$

$\Rightarrow$ In far field, locally plane wave

or w is large, phase difference $\to 0$

Pulsating sphere

$\hat{p}'(r,t) = \hat{p}(r)e^{i\omega t}$

$\Rightarrow \begin{cases} \hat{p}(r) = \frac{A}{r}e^{-ikr} \\ i\omega\rho_0\hat{u}(r) = A\left(\frac{1}{r^2} + \frac{ik}{r}\right)e^{-ikr} \end{cases}$

代入速度边界条件，得 (B.C. 脉动球源表面速度)

$\hat{p}(r) = \frac{a}{r}\frac{ika}{1+ika}\rho_0 c\hat{u}_a e^{-ik(r-a)}$

① $ka \gg 1 \Rightarrow \hat{p}(r) = \frac{a}{r}\rho_0 c\hat{u}_a e^{-ik(r-a)}$

② $ka \ll 1 \Rightarrow \hat{p}(r) = \frac{i\omega a^2}{r}\rho_0\hat{u}_a e^{-ikr}$

$\dot{V}(t) = 4\pi a^2 u_a(t) \Rightarrow p'(r,t) = \frac{\rho_0 \ddot{V}(t-r/c)}{4\pi r}$

monopole

Oscillating rigid sphere

$\frac{\partial}{\partial x_1}\left(\frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2} - \nabla^2\phi\right) = 0$

$\Rightarrow \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\left(\frac{\partial\phi}{\partial x_1}\right) - \nabla^2\left(\frac{\partial\phi}{\partial x_1}\right) = 0$

$\Rightarrow p'(x,t) = \frac{\partial}{\partial x_1}\left[\frac{f(r-ct)}{r}\right]$

代入 $i\omega\rho_0 u = -\frac{\partial p'}{\partial r}$, $\frac{\partial}{\partial x_1} = \frac{\partial r}{\partial x_1}\frac{\partial}{\partial r}$

如果给出振动速度 $u = U\cos\theta\, e^{i\omega t}$

$\Rightarrow p'(r,\theta,t) = \frac{i\omega\rho_0 U a^3\cos\theta}{2(1+ika)-(ka)^2}\left(\frac{1}{r^2} + \frac{ik}{r}\right)e^{i[\omega t - k(r-a)]}$

① $ka \gg 1$, $p' = \rho_0 c U\cos\theta\frac{a}{r}e^{i[\omega t - k(r-a)]}$

$= \frac{a}{r}p'(a,\theta,t-\frac{r-a}{c})$

Radiated sound power $= \frac{2}{3}\pi\rho_0 c U^2 a^2$

$dA = 2\pi(R\sin\theta)(R\,d\theta)$

Drag $= -\int_0^\pi p'\cos\theta\, dS = -\frac{4}{3}\pi\rho_0 c U a^2 e^{i\omega t}$

work $= \overline{DU} = \frac{2}{3}\pi\rho_0 U^2 a^2 =$ Radiated power

② $ka \ll 1$

$p' = -\frac{1}{2}(ka)^2\rho_0 c U\cos\theta\frac{a}{r}e^{i(\omega t - kr)}\left(1 - \frac{i}{kr}\right)$

near field, $kr \ll 1 \Rightarrow 1 - \frac{i}{kr} \approx -\frac{i}{kr}$

$p' = \frac{1}{2}\rho_0 i\omega U\cos\theta\frac{a^3}{r^2}e^{i(\omega t - kr)}$

far field, $kr \gg 1$

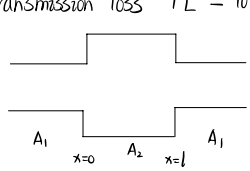
$p' = -\frac{1}{2}(ka)^2\rho_0 c U\cos\theta\frac{a}{r}e^{i(\omega t - kr)}$

Radiated power $= \frac{1}{6}\pi\rho_0 c U^2 a^2 (ka)^4$

$\Re\left(Ae^{i\omega t}\right)\cdot\Re\left(Be^{i\omega t}\right) = \frac{1}{2}\Re(AB^*)$

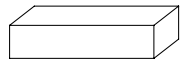
Wave in pipe

transmission loss $TL = 10\log_{10}\left(\frac{A_1|I|^2}{A_2|T^2|}\right)$



$\Re = \frac{\left(\frac{A_1}{A_2} - \frac{A_2}{A_1}\right)i\sin\frac{\omega l}{c}}{2\cos\frac{\omega l}{c} + \left(\frac{A_1}{A_2} + \frac{A_2}{A_1}\right)i\sin\frac{\omega l}{c}}I$, $T = \frac{2I e^{i\omega l/c}}{2\cos\frac{\omega l}{c} + \left(\frac{A_1}{A_2} + \frac{A_2}{A_1}\right)i\sin\frac{\omega l}{c}}$

Rectangular duct



$\omega_c = \pi c\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)^{1/2}$

circular pipe



$\frac{\partial^2 p'}{\partial x^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p'}{\partial r}\right) + \frac{1}{r^2}\frac{\partial p'}{\partial\theta} - \frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} = 0$

$\Rightarrow p' = J_n(z_{mn}r/a)\left(A_{mn}e^{-ik_{mn}x_3} + Be^{ik_{mn}x_3}\right)e^{i(\omega t - n\theta)}$

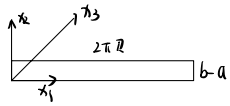
$k_{mn} = \sqrt{k^2 - z_{mn}^2/a^2}$

Plane wave $m=1, n=0$ $\quad (z_{mn})_{min} = 1.841$ $\quad (m=1, n=1)$

	n = 0	n = 1	n = 2
m = 1	0.000	1.841	3.054
m = 2	3.832	5.331	6.706
m = 3	7.016	8.536	9.969

Duct of thin annulus

展开



$\frac{b}{a} \approx 1$

$R = \frac{a+b}{2}$

The periodic condition

$h(x_3) = A_{mn}e^{-ik_{mn}x_3} + B_{mn}e^{ik_{mn}x_3}$

where $k_{mn} = \left[\frac{\omega^2}{c^2} - \frac{n^2}{R^2} - \left(\frac{m\pi}{b-a}\right)^2\right]^{\frac{1}{2}}$

If $\frac{\omega}{c} < \frac{n}{R}$, k_{mn}^2 will be negative for all m

Pipes of varying cross-section

$A\frac{\partial\rho'}{\partial t} = -\rho_0\frac{\partial}{\partial x}(uA)$ $\xrightarrow{\frac{\partial}{\partial t}}$

$\rho_0\frac{\partial u}{\partial t} = -\frac{\partial p'}{\partial x}$ $\xrightarrow{\cdot A,\ \text{div}}$

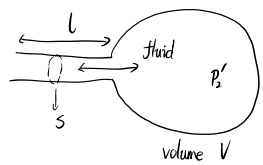
$\frac{A}{c^2}\frac{\partial^2 p'}{\partial t^2} = \frac{\partial}{\partial x}\left(A\frac{\partial p'}{\partial x}\right)$

For $A(x) = A_0 e^{\alpha x}$

$\Rightarrow p'(x,t) = e^{-\alpha x/2}\left[Ae^{i(\omega t - kx)} + Be^{i(\omega t + kx)}\right]$

where $k = \sqrt{\omega^2/c^2 - \alpha^2/4}$

Helmholtz Resonator



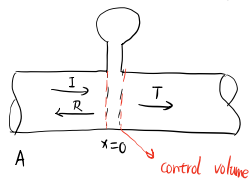
$V\frac{\partial\rho'}{\partial t} = Q \Rightarrow \rho' = \frac{Q}{i\omega V}$ for harmonic waves

$p_2' = c^2\rho' = \frac{c^2 Q}{i\omega V}$

$(p_1' - p_2')S = \rho_0 Sl\frac{\partial u}{\partial t}$, $Q = \rho_0 Su$

$\Rightarrow p_1' = \left(\frac{c^2}{i\omega V} + \frac{i\omega l}{S}\right)Q = \left(\omega^2 - \frac{c^2 S}{Vl}\right)\frac{ilQ}{\omega S}$

$\Rightarrow \omega = \sqrt{\frac{c^2 S}{Vl}}$

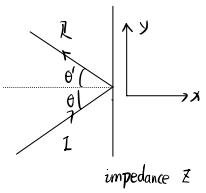


$p_1' = (I+R)e^{i\omega t} = Te^{i\omega t} = \left(\frac{c^2}{i\omega V} + \frac{i\omega l}{S}\right)Q$

Mass flow, $\rho_0\frac{A}{\rho_0 c}(I-R)e^{i\omega t} = Q + \rho_0\frac{AT}{\rho_0 c}e^{i\omega t}$

$\Rightarrow T = \frac{I}{1 + \frac{1}{2A}\left(\frac{c}{i\omega V} + \frac{i\omega l}{cS}\right)}$

Sound wave at interfaces



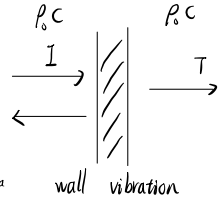
$$\mathcal{R} = \frac{Z - \rho_0 c/\cos\theta}{Z + \rho_0 c/\cos\theta} I$$

If $\mathcal{R} \approx I \Rightarrow$ SPL ↑ 6dB

$I^* \mathcal{R} - I \mathcal{R}^* =$ pure imaginary

The absorption coefficient

$$\alpha = 1 - \left|\frac{\mathcal{R}}{I}\right|^2 - \left|\frac{T}{I}\right|^2$$



By $p_1' = \cdots$

$p_2' = \cdots$

Jump condition

$p_1' - p_2' = (1 + \mathcal{R} - T) e^{i\omega t} = m \frac{\partial v}{\partial t}$ (牛二)

net force; mass per unit area

$$V = \frac{I - \mathcal{R}}{\rho_0 c} = \frac{T}{\rho_0 c} \quad \text{(continuity of velocity)}$$

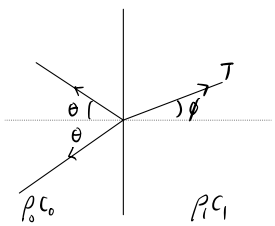
$$\Rightarrow \mathcal{R} = \frac{i\omega m I}{2\rho_0 c + i\omega m} \qquad T = \frac{2\rho_0 c I}{2\rho_0 c + i\omega m}$$

$$V = \frac{2I}{2\rho_0 c + i\omega m}$$

① $\omega m \gg \rho_0 c \Rightarrow |T/I| \approx 0$, $|\mathcal{R}/I| \approx 1$

② $\omega m \ll \rho_0 c \Rightarrow |T/I| \approx 1$, $|\mathcal{R}/I| \approx 0$

Snell's Law



$$\frac{\sin\theta}{c_0} = \frac{\sin\phi}{c_1}$$

$$\mathcal{R} = \frac{\frac{\rho_1 c_1}{\cos\phi} - \frac{\rho_0 c_0}{\cos\theta}}{\frac{\rho_1 c_1}{\cos\phi} + \frac{\rho_0 c_0}{\cos\theta}} I, \quad T = \frac{\frac{2\rho_1 c_1}{\cos\phi}}{\frac{\rho_1 c_1}{\cos\phi} + \frac{\rho_0 c_0}{\cos\theta}} I$$

Ray Theory

$$\frac{\sin\theta}{c(x)} = \text{constant}$$

$$y' = \frac{dy}{dx} = \tan\theta, \quad \sin^2\theta = \frac{\tan^2\theta}{1 + \tan^2\theta}$$

① If $c(x) = \alpha x + \beta$ (线性变化)

$$\Rightarrow y = \frac{1}{\alpha \tan\theta_0}\left(\mp\sqrt{\beta^2 - m^2\theta_0 (\alpha x + \beta)^2} \pm \beta\cos\theta_0\right)$$

$$\left(x + \frac{\beta}{\alpha}\right)^2 + \left(y - \frac{\beta}{\alpha\tan\theta_0}\right)^2 = \frac{\beta^2}{\alpha^2 \sin^2\theta_0}$$

轨迹方程是一个圆

Ray tube $\quad \overline{p'u}A = \text{constant}$

$$\Rightarrow \frac{\overline{p'^2}A}{\rho_0 c} = \text{constant}$$

Uniform media, 2-D: $p'_{rms} \propto \frac{1}{\sqrt{r}}$, 3-D: $p'_{rms} \propto \frac{1}{r}$

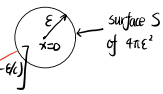
Sound travels further at night.

Sources of Sound

near a singularity $\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{f(t - r/c)}{r} = 0$

$$\int_V \nabla^2\left[\frac{f(t-r/c)}{r}\right] dV = \int_S \frac{\partial}{\partial n}\left[\frac{f(t-r/c)}{r}\right] dS$$

divergence theorem



$$= 4\pi\varepsilon^2\left[-\frac{f(t-\varepsilon/c)}{\varepsilon^2} + \frac{1}{\varepsilon}\frac{\partial f}{\partial r}(t - r/c)\right]$$

$$= -4\pi f(t) \quad \text{as } \varepsilon \to 0$$

$$\Rightarrow \int_V \left[\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{f(t-r/c)}{r}\right] dV \approx 4\pi f(t)$$

$$\int_V 4\pi f(t)\,\delta(x)\,dV$$

$$\Rightarrow \boxed{\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{f(t-r/c)}{r} = 4\pi f(t)\,\delta(x)}$$

$$\boxed{\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = q(x,t)}$$

$$\Rightarrow \boxed{\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\frac{q(y, t - r/c)}{r} = 4\pi\, q(y,t)\,\delta(x-y)}$$

$$\Rightarrow \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)\int_V \frac{q(y, t - |x-y|/c)}{|x-y|}\, d^3y = 4\pi\, q(x,t)$$

$$\Rightarrow p'(x,t) = \int_V \frac{q(y, t - |x-y|/c)}{4\pi|x-y|}\, d^3y$$

Monopole source

Velocity potential $\phi(r,t) = \frac{f(t - r/c)}{r}$

$$\Rightarrow u = \frac{\partial\phi}{\partial r} = -\frac{f(t-r/c)}{r^2} - \frac{\dot f(t-r/c)}{rc}$$

Mass eq. $\quad 4\pi\varepsilon^2\rho_0 u(\varepsilon, t) = m(t)$

monopole source strength 质量注入率的变换率

$$p'(t) = -\rho_0\frac{\partial\phi}{\partial t} = -\rho_0\frac{\dot f(t-r/c)}{r} = \frac{\dot m(t - r/c)}{4\pi r}$$

monopole point source

$$\frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = \dot m(t)\delta(x) \Rightarrow p'(r,t) = \frac{\dot m(t - r/c)}{4\pi r}$$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla_x^2\right)p' = Q(y,t)\,\delta(x-y) \Rightarrow p'(x,t) = \frac{Q(y, t - |x-y|/c)}{4\pi|x-y|}$$

Monopole sources distribution

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p' = q(x,t), \quad q(x,t) = \int q(y,t)\,\delta(x-y)\, d^3y$$

$$\Rightarrow p'(x,t) = \int \frac{q(y, t - |x-y|/c)}{4\pi|x-y|}\, d^3y$$

far field $\quad |x| \gg |y|$

$$(1+\varepsilon)^n = 1 + n\varepsilon + \varepsilon^2$$

$$|x-y| \approx \overline{Ox} - OP = |x| - |y|\cos\theta = |x| - \frac{x\cdot y}{|x|}$$

$1/|x-y| \approx 1/|x|$

$$\Rightarrow p'(x,t) = \frac{1}{4\pi|x|}\int q\left(y, t - \frac{|x|}{c} + \frac{x\cdot y}{|x|c}\right) d^3y$$

compact source $\quad kl \ll 1$

Taylor expansion: $f(x_0 + x) = f(x_0) + x\dot f(x_0)$

$$\Rightarrow t - \frac{|x|}{c} + \frac{x\cdot y}{|x|c} \approx t - \frac{|x|}{c}$$

$$\Rightarrow p'(x,t) = \frac{1}{4\pi|x|}\int q\left(y, t - \frac{|x|}{c}\right) d^3y = \frac{Q(t - |x|/c)}{4\pi|x|}$$

where $Q(t) = \int q(y,t)\, d^3y$

Dipole source

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p' = -\nabla\cdot\left[F(t)\,\delta(x)\right]$$

$$\Rightarrow \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p' = -\frac{\partial}{\partial x_1}\left[F(t)\,\delta(x)\right]$$

$$\Rightarrow p'(x,t) = -\frac{\partial}{\partial x_1}\left[\frac{F(t - r/c)}{4\pi r}\right]$$

代入 $\frac{\partial F}{\partial r} = -\frac{1}{c}\frac{\partial F}{\partial t}$, $\frac{\partial r}{\partial x_1} = \frac{x_1}{r} = \cos\theta$

$$\Rightarrow p'(x,t) = \frac{\cos\theta}{4\pi}\left[\frac{1}{r^2}F(t - r/c) + \frac{1}{rc}\frac{\partial F}{\partial t}(t - r/c)\right]$$

near field $\sim O\left(\frac{F}{r^2}\right)$ $\quad$ far field $\sim O\left(\frac{\omega F}{rc}\right)$

Dipole source distribution

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p' = -\nabla\cdot f(x,t) = -\frac{\partial f_i}{\partial x_i}(x,t)$$

where $\int f(x,t)\, dV = F(t) \neq 0$

The solution to $\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)S_i = f_i(x,t)$ is

$$S_i(x,t) = \int \frac{f_i(y, t - |x-y|/c)}{4\pi|x-y|}\, d^3y$$

$$\Rightarrow p'(x,t) = -\frac{\partial}{\partial x_i}\int \frac{f_i(y, t - |x-y|/c)}{4\pi|x-y|}\, d^3y$$

The far field form $\quad |x| \gg |y|$

代入 $\frac{\partial}{\partial x_i} = \frac{\partial\tau}{\partial x_i}\frac{\partial}{\partial\tau} = \frac{\partial|x|}{\partial x_i}\frac{\partial\tau}{\partial|x|}\frac{\partial}{\partial\tau}$ ✱

$\frac{\partial|x|}{\partial x_i} = \frac{x_i}{|x|}$, $\frac{\partial\tau}{\partial|x|} = -\frac{1}{c}$

$$p'(x,t) = \frac{x_i}{4\pi|x|^2 c}\int \frac{\partial f_i}{\partial t}\left(y, t - \frac{|x|}{c} + \frac{x\cdot y}{|x|c}\right) d^3y$$

Compact source $\quad kl \ll 1$

$$p'(x,t) = \frac{x_i}{4\pi|x|^2 c}\frac{\partial}{\partial t}\int f_i\left(y, t - \frac{|x|}{c}\right) d^3y$$

$$= \frac{x\cdot\dot F(t - |x|/c)}{4\pi|x|^2 c} \quad \text{where } F(t) = \int f(y,t)\, d^3y$$

Combustion noise

$$\frac{\partial^2\rho'}{\partial t^2} - \nabla^2 p' = 0$$

$$d\rho = \frac{\partial\rho}{\partial p}\Big|_s dp + \frac{\partial\rho}{\partial s}\Big|_p ds$$

$$= \frac{1}{c^2}dp + \frac{\partial\rho}{\partial s}\Big|_p ds$$

$$s - s_0 = c_v \ln(P/P_0) - c_p \ln(\rho/\rho_0)$$

$\frac{\partial}{\partial s}$

$$\frac{\partial\rho}{\partial s}\Big|_p = -\frac{\rho}{c_p}$$

$$\rho T\frac{ds}{dt} = w(x,t)$$

$$\Rightarrow \frac{d\rho}{dt} = \frac{1}{c^2}\frac{dP}{dt} - \frac{\gamma - 1}{c^2}w$$

$$\Rightarrow \frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = \frac{\gamma - 1}{c^2}\frac{\partial w}{\partial t}$$

$$\Rightarrow p'(x,t) = \frac{\gamma - 1}{4\pi c^2}\frac{\partial}{\partial t}\int \frac{w(y, t - |x-y|/c)}{|x-y|}\, d^3y$$

linear mass creation & external force

$$\frac{\partial^2\rho'}{\partial t^2} - \nabla^2 p' = \frac{\partial\dot m}{\partial t} - \nabla\cdot f$$

But $\quad p' \neq c^2\rho'$ $\quad \rho = \beta\rho_m + (1-\beta)\rho_f$

$$\cdots \Rightarrow \frac{1}{c^2}\frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = \rho_0\frac{\partial^2\beta}{\partial t^2} - \nabla\cdot f$$

For oscillating sphere

$$p'(r,\theta,t) = -\frac{i\omega\rho_0 U a^3\cos\theta}{2(1 + ika) - (ka)^2}\frac{\partial}{\partial r}\left\{\frac{e^{i[\omega t - k(r-a)]}}{r}\right\}$$

$\frac{\partial}{\partial x_1} = \frac{\partial r}{\partial x_1}\frac{\partial}{\partial r} = \cos\theta\frac{\partial}{\partial r}$

$$p'(x,t) = -\frac{1}{2}i\omega\rho_0 U a^3\frac{\partial}{\partial x_1}\left[\frac{e^{i(\omega t - kr)}}{r}\right]$$

$$-\frac{2}{3}\pi\rho_0 a^3 i\omega U e^{i\omega t} = -\frac{2}{3}\pi\rho_0 a^3\dot U(t)$$

$$F(t) + \underbrace{\rho_0 V\dot U(t)}_{\text{surface displacement}} = \left(\frac{2}{3} + \frac{4}{3}\right)\pi\rho_0 a^3\dot U(t) = \underbrace{2\pi\rho_0 a^3\dot U(t)}_{\text{dipole strength}}$$

Lighthill's acoustic analogy

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = 0 \quad , \quad \frac{\partial}{\partial t}(\rho u_i) = -\frac{\partial}{\partial x_j}(\rho u_i u_j + p'\delta_{ij} - \tau_{ij})$$

where $\rho u_i u_j$ is Reynold's stress

$$P_{ij} = p'\delta_{ij} - \tau_{ij}$$

$$\boxed{\frac{\partial^2 \rho'}{\partial t^2} - c^2 \nabla^2 \rho' = \frac{\partial^2 T_{ij}}{\partial x_i \partial x_j}}$$

where $T_{ij} = \rho u_i u_j + P_{ij} - c^2 \rho' \delta_{ij}$

The solution to $\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right) S = T_{ij}(x,t)$ is

$$S(x,t) = \int \frac{T_{ij}(y, t - |x-y|/c)}{4\pi |x-y|} d^3y$$

$$\Rightarrow \quad \boxed{\rho'(x,t) = \frac{\partial^2}{\partial x_i \partial x_j} \int \frac{T_{ij}(y, t-|x-y|/c)}{4\pi c^2 |x-y|} d^3y}$$

In linear invisid flow $\longrightarrow T_{ij} = 0$

- $\rho u_i u_j \sim O^2(u)$, negligible
- $\tau_{ij} = 0 \rightarrow P_{ij} = p'\delta_{ij}$, $p' = c^2 \rho' \Rightarrow T_{ij} = \rho u_i u_j$

Jet noise

velocity U , diameter D

$\Rightarrow$ frequency $\omega = U/D$, Time scale $\sim D/U$

Wavelength $\lambda \sim cT = \frac{cD}{U} = D M^{-1}$

Helmhotz number $kD = \frac{\omega}{c} D = M$

In far field, $\frac{\partial}{\partial x_i} = \frac{\partial |x|}{\partial x_i}\frac{\partial t}{\partial |x|}\frac{\partial}{\partial t} = -\frac{x_i}{|x|}\frac{1}{c}\frac{\partial}{\partial t} \sim \frac{1}{c}\frac{U}{D} = \frac{M}{D}$

And, $T_{ij} \sim \rho_0 U^2$, $\int_V d^3y \sim D^3$

$$\Rightarrow \quad \rho' \sim \left(\frac{M}{D}\right)^2 \rho_0 U^2 \frac{1}{|x| c^2} D^3 = \rho_0 M^4 \frac{D}{|x|}$$

$$\Rightarrow \quad \overline{\rho'^2} \sim \rho_0^2 M^8 \frac{D^2}{|x|^2} \quad , \quad 8\text{th-pow law}$$

for low Mach number

- In jet $p' \sim \rho_0 U^2$ (hydrodynamic)
- In far field $p' \sim \rho_0 U^2 \times \frac{U^2}{c^2}\frac{D}{|x|}$ (acoustic)

For high Mach number,

$$\rho'(x,t) = \frac{1}{4\pi c r}\frac{\partial^2}{\partial r^2}\int T_{rr}(y,\tau)\, dS_r d\tau$$

$T_{rr} \sim \rho_0 U^2$, $\frac{\partial}{\partial r} \sim \frac{1}{D}$, $\int_S dS_r \sim D^2$, $\int_\tau d\tau \sim \frac{D}{U}$

$$\Rightarrow \quad \rho' \sim \frac{1}{Cr}\frac{1}{D^2}\rho_0 U^2 D^2 \frac{D}{U} = \rho_0 M \frac{D}{r}$$

$$\Rightarrow \quad \overline{\rho'^2} \sim \rho_0^2 M^2 \frac{D^2}{r^2}$$

The sound of foreign bodies

$$\frac{\partial^2}{\partial x_i \partial x_j}\int_{V_0}\left[\frac{T_{ij}}{r}\right] dV = -\frac{\partial}{\partial x_j}\int_S n_i\left[\frac{\rho u_i u_j + P_{ij}}{r}\right] dS + \frac{\partial}{\partial t}\int_S \left[\frac{\rho u \cdot n}{r}\right] dS$$

dipole monopole

$$\rho'(x,t) = \frac{1}{4\pi c^2}\frac{\partial^2}{\partial x_i \partial x_j}\int_V \left[\frac{T_{ij}}{r}\right] dV - \frac{1}{4\pi c^2}\frac{\partial}{\partial x_j}\int_S n_i\left[\frac{\rho u_i u_j + P_{ij}}{r}\right] dS + \frac{1}{4\pi c^2}\frac{\partial}{\partial t}\int_S \left[\frac{\rho u \cdot n}{r}\right] dS$$

- If foreign body stationary and rigid $\longrightarrow$ $n_i(\rho u_i u_j) = 0$, $u \cdot n = 0$ on S

- The sound field by foreign body in inviscid fluid ($\tau_{ij} = 0$) , $P_{ij} = p'\delta_{ij}$

$$-\frac{1}{4\pi c^2}\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right] dS$$

- For a compact body , τ variation $\longrightarrow 0$

$$-\frac{1}{4\pi c^2}\frac{\partial}{\partial x_i}\left[\frac{F_i}{r}\right] \quad \text{with} \quad F_i = \int_S n_i p' dS$$

(force applied by body to fluid)

Sound by compact rigid bodies in turbulent flow

$\Leftarrow$ unsteady force acting on bodies $\quad F \sim \rho_0 U^2 D^2$

Use $\boxed{\partial/\partial x_i \sim M/D} \Rightarrow \rho' \sim \rho_0 M^3 \frac{D}{r}$ and $\overline{\rho'^2} \sim \rho_0^2 M^6 \frac{D^2}{r^2}$

- Body forces generate sound more efficient than low Mach number free turbulence by M^{-2}

$\frac{1}{4\pi c^2}\frac{\partial}{\partial t}\left[\frac{Q}{r}\right]$ with $Q = \int_S \rho u_n\, ds$ Use $\frac{\partial}{\partial t} \sim U/D$, $Q \sim \rho_0 U D^2$

$$\Rightarrow \quad \rho' \sim \rho_0 M^2 \frac{D}{r} \quad \text{and} \quad \overline{\rho'^2} \sim \rho_0^2 M^4 \frac{D^2}{r^2}$$

- Sound generation ($M \ll 1$) : mass (volume) flux >> body force >> turbulent flow by a factor of M^{-2} and M^{-4}

FWH equation

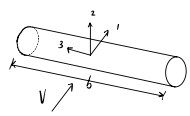
Assumption : 忽略湍流 T_{ij} 项, rigid body , inviscid linear motion : $\rho u_i u_j \approx 0$, $p' = c^2 \rho'$

$$\boxed{p'(x,t) = \frac{\partial}{\partial t}\int_S \frac{\rho_0 n \cdot u(y, t-|x-y|/c)}{4\pi |x-y|} dS - \frac{\partial}{\partial x_i}\int_S \frac{n_i p(y, t-|x-y|/c)}{4\pi |x-y|} dS}$$

In far field,

$$\boxed{p'(x,t) = \frac{1}{4\pi |x|}\frac{\partial}{\partial t}\int_S \rho_0 n \cdot u\left(y, t - \frac{|x|}{c} + \frac{x \cdot y}{|x| c}\right) dS + \frac{x_i}{4\pi |x|^2 c}\frac{\partial}{\partial t}\int_S n_i p\left(y, t - \frac{|x|}{c} + \frac{x \cdot y}{|x| c}\right) dS}$$

Case 1 : Rigid body at rest $\quad u \cdot n = 0$



far field : $p'(x,t) = \frac{x_i}{4\pi |x|^2 c}\frac{\partial}{\partial t}\int_S n_i p\left(y, t - \frac{|x|}{c}\right) dS$

Frequency $\propto \frac{V}{D} = St\frac{2\pi V}{D}$ with $St = \frac{fD}{V} \sim 0.2$

Note $kD = St\frac{2\pi V}{D}\frac{1}{c}D \sim St\, 2\pi M \rightarrow$ Compactness means low Mach number

Let unsteady force / unit length $= f(y_3, t)$ in 2-direction

$\Rightarrow$ $f_{rms} = C_L \frac{1}{2}\rho_0 V^2 D$ and $\frac{\partial f}{\partial t} = iSt\frac{2\pi V}{D} f$

f_{rms} 单位展长的升力 ; lift coefficient ; $= i\omega f \sim i\, St\frac{2\pi V}{D} f$

Use $p'(x,t) = \frac{x_i}{4\pi |x|^2 c}\frac{\partial}{\partial t}\int_S n_i p\left(y, t - \frac{|x|}{c}\right) dS$ 表面压力的面积分表现为电线受到的升力

$$\Rightarrow \quad p'(x,t) = \frac{x_2}{4\pi |x|^2 c} i\, St\frac{2\pi V}{D}\int_b^b f(y_3, t - \frac{|x|}{c})\, dy_3$$

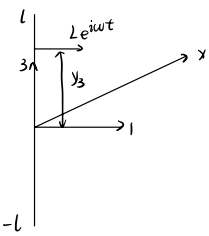
Autocorrelation : $F(\xi) = \overline{f(y_3, t)\, f(y_3 + \xi, t)}$ $\quad$ 特征长度 $l\overline{f^2} = \int_{-\infty}^{\infty} F(\xi)\, d\xi$

$$\Rightarrow \quad \overline{p'^2} = \left(\frac{x_2 St\, V}{2|x|^2 c\, D}\right)^2 \int_b^b l\overline{f^2}\, dy_3 = \left(\frac{x_2 St\, V}{2|x|^2 c\, D}\right)^2 2bl\overline{f^2}$$

Use $f_{rms} = C_L \frac{1}{2}\rho_0 V^2 D$ $\Rightarrow$ $\overline{p'^2} = \left(\frac{x_2 St\, V}{2|x|^2 c\, D}\right)^2 2bl\left(C_L \frac{1}{2}\rho_0 V^2 D\right)^2$

$$= \frac{x_2^2}{8|x|^4 c^2} St^2\, bl\, C_L^2 \rho_0^2 V^6$$

An example for



(在后面附加页)

Case 2 A compact body producing a net monopole

For $|x| \gg |y|$ and a compact body, Eq (6.58) reduce to

$$p'(x,t) = \frac{1}{4\pi|x|}\frac{\partial}{\partial t}\int_S \rho_0 n\cdot u(y, t-\tfrac{|x|}{c})\,dS + \frac{x_i}{4\pi|x|^2 c}\frac{\partial}{\partial t}\int_S n_i p(y, t-\tfrac{|x|}{c})\,dS$$

$$= \frac{\dot{Q}(t-|x|/c)}{4\pi|x|} + \frac{x\cdot \dot{F}(t-|x|/c)}{4\pi|x|^2 c}$$

where $Q(t) = \rho_0 \int_S n\cdot u\,dS$, $F(t) = \int_S n\,p\,dS$

量级比较 $Q \sim \rho_0 V D^2$ $\quad F \sim \rho_0 U^2 D^2$

$$\Rightarrow \frac{\dot{Q}}{4\pi|x|} \sim \frac{\rho_0 \omega V D^2}{|x|} \qquad \frac{x\cdot\dot{F}}{4\pi|x|^2 c} \sim \frac{\rho_0 \omega^2 V D^3}{|x| c} = \frac{\rho_0 \omega V D^2}{|x|}(kD)$$

small for compact

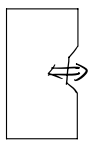
$\Rightarrow$ neglect dipole

Hence for a compact body producing a net monopole, the far-field pressure.

$$p'(x,t) = \frac{\dot{Q}(t-|x|/c)}{4\pi|x|} \quad \text{where} \quad Q(t) = \rho_0 \int_S n\cdot u\,dS$$

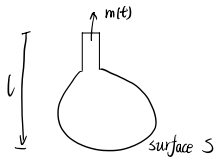
a loudspeaker $\int_S n\cdot u\,dS = \dot{V}(t)$

$$\Rightarrow p'(x,t) = \rho_0 \frac{\ddot{V}(t-|x|/c)}{4\pi|x|}$$



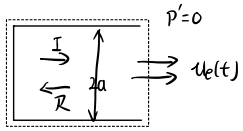
Helmholtz resonator

$$p'(x,t) = \frac{\dot{m}(t-|x|/c)}{4\pi|x|} \quad \text{where} \quad m(t) = \rho_0 \int_S n\cdot u\,dS$$



Open end of organ pipes

ka << 1 with a the radius



For x in far field,

$$p'(x,t) = \frac{\rho_0 \pi a^2 \dot{u}_e(t-|x|/c)}{4\pi|x|}$$

Radiated sound power $= \frac{\overline{p'^2}}{\rho_0 c} 4\pi|x|^2 = \frac{\rho_0 \omega \pi a^2 u_e^2}{4c}$

B.C. at open end (x=0)

(1) $p'=0 \Rightarrow I+R=0$ (2) $u_e = \frac{I-R}{\rho_0 c} = \frac{2I}{\rho_0 c} \Rightarrow u_i = \frac{I}{\rho_0 c} = \frac{1}{2}u_e$

Incident sound power $= \overline{(\frac{1}{2}u_e)^2}\,\rho_0 c \pi a^2 = \frac{\rho_0 c \pi a^2 \overline{u_e^2}}{4}$

$$\Rightarrow \frac{\text{Radiated sound power}}{\text{Incident sound power}} = (ka)^2 << 1$$

Case 3: A moving rigid compact body

$$p'(x,t) = \frac{\rho_0}{4\pi|x|}\frac{x\cdot\dot{U}(t-|x|/c)V}{|x|c} + \frac{x\cdot\dot{F}(t-|x|/c)}{4\pi|x|^2 c}$$

$$= \frac{x\cdot[\rho_0 V\dot{U}(t-|x|/c) + \dot{F}(t-|x|/c)]}{4\pi|x|^2 c}$$

$$= \frac{-\nabla\cdot[\rho_0 V\dot{U}(t-|x|/c) + F(t-|x|/c)]}{4\pi|x|}$$

$\frac{\partial}{\partial x_i} = -\frac{1}{c}\frac{x_i}{|x|}\frac{\partial}{\partial t}$

- a moving rigid compact body $\longrightarrow$ dipole sound field

Dipole strength $= F(t) + \rho_0 V\dot{U}(t)$

force exerted on fluid by body; rate of dissipated momentum

- primarily due to the unsteady force the body exerts on fluid + additional term due to the inertia of displaced fluid.

Note: If the only force on the body is due to the fluid

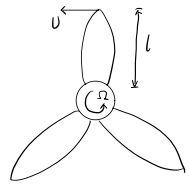
Force on body $= -F(t) = m\dot{U}(t)$

Net dipole strength $= (-m+\rho_0 V)\dot{U}(t)$ (m: mass of the body)

Consider a propeller or fan with n blades of length l, angular velocity Ω

the speed $U = \Omega l$

sound of frequency $\omega = n\Omega$ (要乘上桨叶数)



compactness ratio $= \frac{n\Omega l}{c} = nM << 1$ for compact propeller

Helmholtz number $= \frac{\omega}{c} l$

$\downarrow$

tip Mach number

$\bar{F} = \bar{f} l \sim (C_D \rho_0 U^2 l) l$

Unsteady force $F \sim C_D \rho_0 U^2 l^2$

Far-field sound $p' \sim \frac{C_D}{|x|c}\left(\frac{nU}{l}\right)\rho_0 U^2 l^2 = C_D n \rho_0 \frac{U^3}{c}\frac{l}{r}$

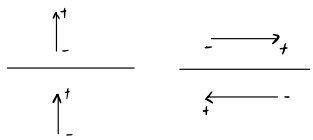
The reciprocal theorem

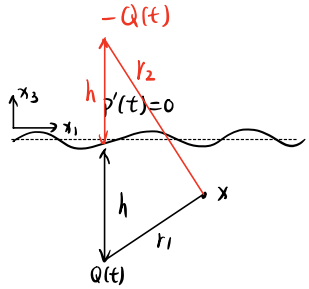
$$\phi_1(x_1) = Q\phi_2(x_0) + T_{ij}\frac{\partial^2 \phi_2(x_1)}{\partial x_0 \partial x_j}$$

- The field at x_2 due to a point monopole at x_0 + a point quadrupole at
= the field at x_0 + double divergence of the field at x_1 due to a monopole at x_2

Plane free surface

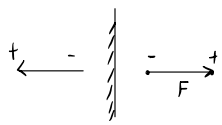
asymmetric

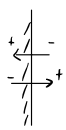




Plane rigid surface

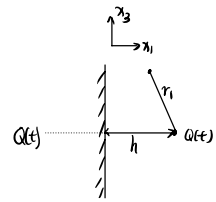
Symmetric

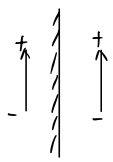
(ii) 



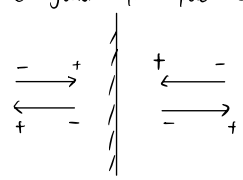
cancel $\Rightarrow$ no sound

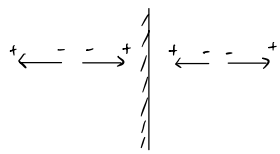
if h = 0



(iii) h=0  $\Rightarrow$ double

(iv) The field quadrupole sources near a hard surface

 $\Rightarrow$ cancel

 $\Rightarrow$ double

Kirchhoff's theorem for plane surfaces

$$\frac{\partial^2}{\partial x_i \partial x_j}\int_{V_0}\left[\frac{T_{ij}}{r}\right]dV = -\frac{\partial}{\partial x_j}\int_S n_i\left[\frac{\rho u_i u_j + P_{ij}}{r}\right]dS + \frac{\partial}{\partial t}\int_S\left[\frac{\rho u\cdot n}{r}\right]dS$$

Linear condition: $T_{ij} = \rho_0 v_i v_j \sim 0$, $P_{ij} = p'\delta_{ij}$, $\rho v_i = \rho_0 u_i$

$$\Rightarrow \quad 0 = -\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS + \rho_0\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS$$

Under linearity constraints, Lighthill's acoustic analogy → a bounded region

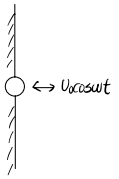
$$\Rightarrow \rho'(x,t) = -\frac{1}{4\pi c^2}\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS + \frac{\rho_0}{4\pi c^2}\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS$$

For plane control surface

$$\rho'(x,t) = \frac{\rho_0}{2\pi c^2}\frac{\partial}{\partial t}\int_S\left[\frac{u\cdot n}{r}\right]dS = -\frac{1}{2\pi c^2}\frac{\partial}{\partial x_i}\int_S n_i\left[\frac{p'}{r}\right]dS$$

for far field $r = |x| - \frac{x\cdot y}{|x|}$

Example: For compact sphere, $|x-y| \approx |x|$



$$p'(x,t) = \frac{\rho_0\pi a^2}{2\pi|x|}\frac{d}{dt}\left(U_0\cos[\omega(t-|x|/c)]\right)$$

$$= -\frac{\rho_0\pi a^2}{2\pi|x|}\omega U_0\sin[\omega(t-|x|/c)]$$

Radiated sound power $= \frac{1}{2}\left(\frac{\rho_0\pi a^2\omega U_0}{2\pi|x|}\right)^2\frac{1}{\rho_0 c}\times 2\pi r^2 = \frac{\pi}{4}\rho_0 c(ka)^2a^2U_0^2$

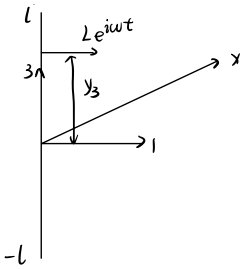
In unbound space, radiated sound power $= \frac{\pi}{6}\rho_0 c(ka)^4a^2U_0^2$

The plane increases the power output by a factor $\frac{3}{2}(ka)^{-2}$

Appendix

Example 6.5

in the x_1-direction on the wire of strength $Le^{i\omega t}$ per unit length of wire



$$p'(x,t) = \frac{\partial}{\partial x_1}\int_{-l}^{l}\frac{Le^{i(\omega t-k|x-y|)}}{4\pi|x-y|}dy_3$$

with $y = (0,0,y_3)$

far-field $|x-y| \approx |x| - \frac{x\cdot y}{|x|} = r - \frac{x_3y_3}{r}$

since $y_1 = y_2 = 0$

$$p'(x,t) = \frac{\partial}{\partial x_1}\left[\frac{Le^{i(\omega t-kr)}}{4\pi r}\int_{-l}^{l}e^{ikx_3y_3/r}dy_3\right]$$

$$= \frac{\partial}{\partial x_1}\left\{\frac{Le^{i(\omega t-kr)}}{4\pi r}\left[\frac{e^{ikx_3y_3/r}}{ikx_3/r}\right]\Bigg|_{-l}^{l}\right\}$$

$$= \frac{\partial}{\partial x_1}\left[\frac{Le^{i(\omega t-kr)}}{4\pi r}\frac{2\sin(klx_3/r)}{kx_3/r}\right]$$

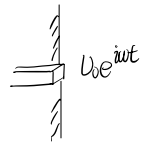
$$\downarrow \quad \frac{\partial}{\partial x_1} = \frac{\partial\tau}{\partial x_1}\frac{\partial}{\partial t} = \frac{\partial r}{\partial x_1}\frac{\partial\tau}{\partial r}\frac{\partial}{\partial t} = \frac{x_1}{r}\left(-\frac{1}{c}\right)i\omega$$

$$= -\frac{ikx}{4\pi r^2}2Ll\frac{\sin(k(x_3/r))}{kl(x_3/r)}e^{i(\omega t-kr)} + O\left(\frac{1}{r^2}\right)$$

near field

If $kl \ll 1$, $\frac{\sin(kl(x_3/r))}{kl(x_3/r)} \to 1$

Example 7.4 Square piston



in the limits of (i) $wa/c \ll 1$ (ii) $wa/c \gg 1$.

For large $|x| = R$

$$p'(x,t) = \frac{i\omega\rho_0U_0}{2\pi R}e^{i(\omega t-kR)}\int_{-a/2}^{a/2}e^{ikx_2y_2/R}dy_2\int_{-a/2}^{a/2}e^{ikx_3y_3/R}dy_3$$

$$= \frac{i\omega\rho_0U_0}{2\pi R}e^{i(\omega t-kR)}\left[\frac{e^{ikx_2y_2/R}}{ikx_2/R}\right]\Bigg|_{-a/2}^{a/2}\left[\frac{e^{ikx_3y_3/R}}{ikx_3/R}\right]\Bigg|_{-a/2}^{a/2}$$

$$= \frac{i\omega\rho_0U_0}{2\pi R}e^{i(\omega t-kR)}\frac{2i\sin\left(\frac{ka\,x_2}{2R}\right)2i\sin\left(\frac{ka\,x_3}{2R}\right)}{(ikx_2/R)(ikx_3/R)}$$

$$= \frac{i\omega\rho_0U_0a^2}{2\pi R}e^{i(\omega t-kR)}\frac{\sin\left(\frac{ka\,x_2}{2R}\right)\sin\left(\frac{ka\,x_3}{2R}\right)}{\frac{ka\,x_2}{2R}\cdot\frac{ka\,x_3}{2R}}$$

For $ka \ll 1$, $p'(x,t) \approx \frac{i\omega\rho_0U_0a^2}{2\pi R}e^{i(\omega t-kR)}$

an omni-directional pattern with source strength dependent on total area a^2

For $ka \gg 1$, $p'(x,t) = 0$ when $\sin\left(\frac{ka\,x_2}{2R}\right)$, $x_2 \neq 0$

$\sin\left(\frac{ka\,x_3}{2R}\right)$, $x_3 \neq 0$

i.e. $\frac{x_2}{R} = \frac{2n\pi}{ka}$ or $\frac{x_3}{R} = \frac{2m\pi}{ka}$ with m and n non-zero integers

→ a pattern of lobes

Oscillating rigid sphere

补充: $p'(x,t) = \frac{\partial}{\partial x_1}\left[\frac{f(r-ct)}{r}\right]$, $\frac{\partial}{\partial x_1} = \frac{\partial r}{\partial x_1}\frac{\partial}{\partial r} = \cos\theta\frac{\partial}{\partial r}$

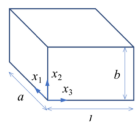
$\Rightarrow p'(x,t) = \cos\theta\frac{\partial}{\partial r}\left[\frac{f(r-ct)}{r}\right]$

代入动量方程 $i\omega\rho_0u = -\frac{\partial}{\partial r}p'$

If harmonic $\Rightarrow f(r-ct) = e^{i\omega(t-r/c)}$

$$\Rightarrow u = -\frac{A\cos\theta}{i\omega\rho_0}\left(\frac{2}{r^3} + \frac{2ik}{r^2} - \frac{k^2}{r}\right)e^{i\omega(t-r/c)}$$

Q2.1



$$\Rightarrow \frac{d^2h(x_3)}{dx_3^2} + \left[\frac{\omega^2}{c^2} - \pi^2\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)\right]h(x_3) = 0$$

$$\Rightarrow h(x_3) = A_{mn}e^{-ik_{mn}x_3} + B_{mn}e^{ik_{mn}x_3}$$

where $k_{mn} = \left[\frac{\omega^2}{c^2} - \pi^2\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)\right]^{\frac{1}{2}}$

Boundary condition: $\left.\frac{dh(x_3)}{dx_3}\right|_{x_3=0} = \left.\frac{dh(x_3)}{dx_3}\right|_{x_3=l} = 0$

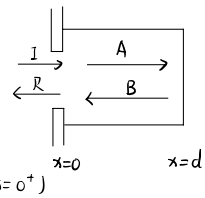
$$\Rightarrow \begin{cases}\left.\frac{dh(x_3)}{dx_3}\right|_{x_3=0} = -ik_{mn}A_{mn} + ik_{mn}B_{mn} = 0\\ \left.\frac{dh(x_3)}{dx_3}\right|_{x_3=l} = -ik_{mn}A_{mn}e^{-ik_{mn}l} + ik_{mn}B_{mn}e^{ik_{mn}l} = 0\end{cases}$$

$$\Rightarrow -k_{mn}A_{mn}\sin(k_{mn}l) - k_{mn}B_{mn}\sin(k_{mn}l) = 0$$

$$\Rightarrow -k_{mn}(A_{mn}+B_{mn})\sin(k_{mn}l) = 0 \Rightarrow \sin(k_{mn}l) = 0 \Rightarrow k_{mn}l = p\pi \quad (p = 0, 1, \ldots)$$

$$\Rightarrow \left[\frac{\omega^2}{c^2} - \pi^2\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)\right]^{\frac{1}{2}} = \frac{p\pi}{l} \Rightarrow \omega = c\pi\left[\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{l}\right)^2\right]^{\frac{1}{2}}$$

Q2.2



B.C. $\begin{cases}\hat{u}(x=d) = 0\\ \hat{p}(x=0^-) - \hat{p}(x=0^+) = \omega l\rho_0 c\,u_h\end{cases}$

where $u_h = \hat{u}(x=0^-)$

And because $u_h = u_n/\alpha$, where $u_h = \hat{u}(x=0^-)$, $u_h = \hat{u}(x=0^+)$

Combine these three equations

Q3.1 $\left(\frac{\partial^2}{\partial t^2} - c^2\nabla^2\right)p' = Ae^{-\sigma x_1^2}\delta(x_2)\delta(x_3)e^{i(\omega t-\alpha x_1)}$

(i) Show that the far-field sound pressure radiated by the wave packet is

$$\frac{A}{4c^2x\sqrt{\pi\sigma}}e^{i(\omega t-kx)}e^{-(k\cos\theta-\alpha)^2/(4\sigma)}$$

where $k = \omega/c$, $x = |\boldsymbol{x}|$, and θ is the polar angle between the observer and the x_1 axis. [10%]

Hint 1: For $x \gg y$, $|\boldsymbol{x}-\boldsymbol{y}| = x(1 - \boldsymbol{x}\cdot\boldsymbol{y}/x^2 + \cdots)$

Hint 2: $\int_{-\infty}^{\infty}e^{-\sigma y^2}e^{i\gamma y}\,dy = \sqrt{\frac{\pi}{\sigma}}e^{-\gamma^2/(4\sigma)}$

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p' = q(\vec{x},t) \quad ①$$

where $q(\vec{x},t) = \frac{A}{c^2}e^{-\sigma x_1^2}\delta(x_2)\delta(x_3)e^{i(\omega t-\alpha x_1)}$

$$p(\vec{x},t) = \int_V\frac{q(\vec{y},t-|\vec{x}-\vec{y}|/c)}{4\pi|\vec{x}-\vec{y}|}d^3y \quad \cdots$$

phase speed

Q 3.4 $\left(\frac{\partial^2}{\partial t^2} - c^2\nabla^2\right)\rho'(\boldsymbol{x},t) = \frac{\gamma-1}{c^2}\frac{\partial w}{\partial t}$

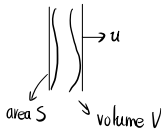
$\Rightarrow \quad \rho'(\vec{x},t) = \frac{\gamma-1}{4\pi|\vec{x}|c^4}\frac{\partial}{\partial t}\int_V w\left(\vec{y}, t-\frac{|\vec{x}|}{c}\right) d^3y$

$\downarrow$

$\frac{\gamma-1}{c^2}\frac{\partial}{\partial t}\int_V w\left(\vec{y}, t-\frac{|\vec{x}|}{c}\right) d^3y = \rho_0\frac{\partial^2}{\partial t^2}\int_V \beta\left(\vec{y}, t-\frac{|\vec{x}|}{c}\right) d^3y$

$\rho'(\vec{x},t) = \frac{\rho_0}{4\pi|\vec{x}|c^2}\frac{\partial^2}{\partial t^2}\int_V \beta\left(\vec{y}, t-\frac{|\vec{x}|}{c}\right) d^3y$

Suppose that the flame is a sheet



$\Rightarrow \quad \frac{\partial}{\partial t}\beta = \frac{uS}{V}$

$\Rightarrow \quad \frac{\partial}{\partial t}\beta \sim \frac{u}{l}$

And because $\int_V d^3y \sim l^3$, $\frac{\partial}{\partial t} \sim \frac{u}{l}$,

$\Rightarrow \quad \rho' \sim \frac{\rho_0}{x c^2}\left(\frac{u}{l}\right)\left(\frac{u}{l}\right) l^3 = \rho_0\left(\frac{l}{x}\right)\left(\frac{u}{c}\right)^2 = \rho_0\left(\frac{l}{x}\right)M^2$

$\Rightarrow \quad \rho' \sim \rho_0\left(\frac{l}{x}\right)M^2$

Supersonic jet

For a supersonic jet, sound field structure $\longleftarrow$ $\tau = t - |x-y|/c$

In far field, $|x-y| \approx |x| - \frac{x\cdot y}{|x|} = r - y_r$

$\frac{\partial^2 T_{ij}}{\partial x_i \partial x_j} = \frac{\partial^2 r}{\partial x_i \partial x_j}\frac{\partial^2 T_{ij}}{\partial r^2} = \frac{\partial^2 T_{rr}}{\partial r^2} \Rightarrow \frac{x_i x_j}{|x|^2}$

The solution simplies to

$\rho'(x,t) = \frac{1}{4\pi c^2 r}\frac{\partial^2}{\partial r^2}\int T_{rr}\left(y, t-r/c+y_r/c\right) d^3y$

$= \frac{1}{4\pi c^2 r}\frac{\partial^2}{\partial r^2}\int T_{rr}(y,\tau)\, dy_r\, dS_r$

- $T_{rr} = \frac{x_i x_j}{|x|^2}T_{ij}$: component of T_{ij} in observer's direction
- $y_r = x\cdot y/|x|$: component of y in direction of x v y
- S_r : surface of y_r = constant

change integration variable $dy_r = c\,d\tau$

$\Rightarrow \quad \rho'(x,t) = \frac{1}{4\pi c r}\frac{\partial^2}{\partial r^2}\int T_{rr}(y,\tau)\, dS_r d\tau$

- $y = (y_s, y_r) = (y_s, c(\tau - t + r))$

$\downarrow$

vector lying in S_r

y_r variation $\sim D/M$ $\qquad$ τ variation $\sim D/U$

negligible for highly supersonic jets

$T_{rr} \sim \rho_0 U^2$ $\quad$ $\frac{\partial}{\partial r} \sim \frac{1}{D}$ $\quad$ $\int_S dS_r \sim D^2$ $\quad$ $\int_\tau d\tau \sim \frac{D}{U}$

$\Rightarrow \quad \rho' \sim \frac{1}{Cr}\frac{1}{D^2}\rho_0 U^2 D^2 \frac{D}{U} = \rho_0 M \frac{D}{r} \quad \Rightarrow \quad \overline{\rho'^2} \sim \rho_0^2 M^2 \frac{D^2}{r^2}$

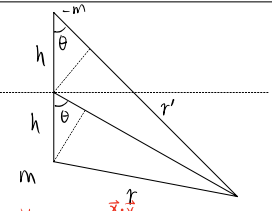
Another effect: No. of eddies heard at $(x,t) \propto M$

$\Rightarrow$ $\overline{\rho'^2} \sim U^2$ in agreement with experiments of rocket noise

Example 7.1

Solution:

The image is a source of strength $-m$ at $x_3 = -h$



$r = \left[x_1^2 + x_2^2 + (x_3-h)^2\right]^{1/2}$

$\approx |x| - \frac{x_3 h}{|x|} = |x| - h\cos\theta$ $\quad$) $|\vec{x}-\vec{y}| \approx |\vec{x}| - \frac{\vec{x}\cdot\vec{y}}{|\vec{x}|}$

$\Rightarrow \quad r' = |x| + h\cos\theta \quad \Rightarrow \quad \frac{x\cdot y}{|x|} = \begin{cases} h\cos\theta & \text{for } m \\ h\cos(\pi-\theta) & \text{for } -m \end{cases}$

Volume outflow rate $\dot{V} = \sqrt{2}\,\dot{m}$ $\longleftarrow$ $m^2 = \overline{\dot{V}^2} = \frac{1}{2}\dot{V}^2$

$\Rightarrow \quad Q = \rho_0\dot{V} = \rho_0\sqrt{2}\,\dot{m} = i\omega\rho_0\sqrt{2}\,m$

$\Rightarrow \quad p'(x,t) = \frac{i\omega\rho_0\sqrt{2}m}{4\pi}\left[\frac{e^{i(\omega t - kr)}}{r} - \frac{e^{i(\omega t - kr')}}{r'}\right]$

$\approx \frac{i\omega\rho_0\sqrt{2}m}{4\pi|x|}e^{i(\omega t - k|x|)}\left(e^{ikh\cos\theta} - e^{-ikh\cos\theta}\right)$

$= -\frac{\omega\rho_0\sqrt{2}m}{2\pi|x|}e^{i(\omega t - k|x|)}\sin(kh\cos\theta)$

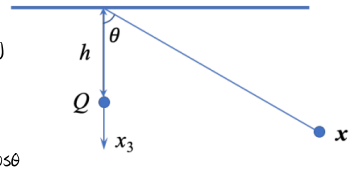
Radiated sound power

$\bar{P} = \int_0^{\frac{\pi}{2}}\frac{\overline{p'^2}}{\rho_0 c}\cdot 2\pi R\sin\theta\, R\, d\theta = \left(\frac{\omega\rho_0 m}{2\pi}\right)^2\frac{1}{\rho_0 c}\int_0^{\frac{\pi}{2}} 2\pi\sin\theta\sin^2(kh\cos\theta)\, d\theta$

$= \frac{\rho_0\omega^2 m^2}{4\pi c}\int_0^{\frac{\pi}{2}}\left[1-\cos(2kh\cos\theta)\right]\sin\theta\, d\theta = \frac{\rho_0\omega^2 m^2}{4\pi c}\left[-\cos\theta + \frac{\sin(2kh\cos\theta)}{2kh}\right]_0^{\frac{\pi}{2}}$

$= \frac{\rho_0\omega^2 m^2}{4\pi c}\left[1 - \frac{\sin(2kh)}{2kh}\right]$

Last Q 3.5



$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2\right)p'(\vec{x},t) = Q(t)\delta(x_3+h) + Q(t)\delta(x_3-h)$

$\Rightarrow \quad p'(\vec{x},t) = \frac{Q(t-|\vec{r}_1|/c)}{4\pi|\vec{r}_1|} + \frac{Q(t-|\vec{r}_2|/c)}{4\pi|\vec{r}_2|}$

For far field $|\vec{r}_1| = |\vec{x}| - h\cos\theta$, $|\vec{r}_2| = |\vec{x}| + h\cos\theta$

$\frac{1}{|\vec{r}_1|} = \frac{1}{|\vec{r}_2|} = \frac{1}{|\vec{x}|}$

$\Rightarrow \quad p'(\vec{x},t) = \frac{Q e^{i\omega t - ik|\vec{x}|}}{4\pi|\vec{x}|}\left(e^{ikh\cos\theta} + e^{-ikh\cos\theta}\right) = \frac{Q\cos(kh\cos\theta)e^{i\omega t - ik|\vec{x}|}}{2\pi|\vec{x}|}$

When $\cos(kh\cos\theta) = 0$ $\Rightarrow$ cancel $\Rightarrow$ $\cos\theta = \frac{\frac{\pi}{2}+n\pi}{kh}$

$\Rightarrow$ Radiated sound power $= \int_{\partial V}\frac{\overline{p'^2}}{\rho_0 c}dA = \int_0^{\frac{\pi}{2}}\frac{\overline{p'^2}}{\rho_0 c}2\pi R\sin\theta\, R\, d\theta$

$= \frac{Q^2}{8\pi\rho_0 c}\left(1 + \frac{\sin(2kh)}{2kh}\right)$

Without boundary $\Rightarrow$ Radiated sound power $= \frac{Q^2}{8\pi\rho_0 c}$